

Answers: Mean Class Size

Suppose a college is offering five sections of an introductory statistics course. The enrollments for the sections are:

Section	1	2	3	4	5
Enrollment	200	35	35	20	10

a) What are the observational units? What is the variable?

The observational units are the **5 sections** of the course. The variable is the **enrollment** (number of students) in each section.

b) Calculate the mean enrollment per section.

$$\bar{x} = \frac{200 + 35 + 35 + 20 + 10}{5} = \frac{300}{5} = 60 \text{ students}$$

c) Mean class size from the students' point of view

Now there are 300 observational units (students). Each student reports the size of their own class:

Section	Students reporting	Value each reports	Total
1	200	200	40,000
2	35	35	1,225
3	35	35	1,225
4	20	20	400
5	10	10	100
All	300		42,950

$$\bar{x} = \frac{42,950}{300} \approx 143.2 \text{ students}$$

The mean is much bigger from the students' point of view. Most students (200 of 300) are in the large section, so a typical student is in a big class, even though the typical section is much smaller. Which mean you get depends on what the observational units are.